

# Smart Cradle: A Secured Assistance and Monitoring system for baby using IoT

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**Abstract**— Nowadays, people seem to not care about their work and may not always be able to take care of their children. Working mothers facing difficulties at times to take care of children, hence we aim to create a IoT based baby care system to solve the challenges faced by caregivers in ensuring the safety and well-being of infants. Based on existing research and technology, the new system is created has a monitoring solution that can monitor critical environmental conditions for baby care. Our goal is to examine the effectiveness of integrating Proteus software, Arduino boards, temperature and humidity sensors, GSM modules, virtual terminals, and DC motors in creating a security monitoring system. Through evaluation and analysis, we attempt to measure the effectiveness and efficiency of the baby monitoring system in providing real-time monitoring, remote access, and timely alarm reports. Our research approach, which uses a combination of hardware and software, will include modelling, simulation, and optimization to achieve the desired results. The system promotes advanced infant care technology by providing caregivers with powerful and reliable solutions to monitor and secure babies in many locations.

**Keywords**— IoT, Proteus Simulation, Caretaker challenges, Arduino boards.

## I. INTRODUCTION

Modern technology has impacted many facets of daily life, providing previously unheard-of convenience and increased security. Among these developments, the incorporation of Internet of Things (IoT) technology has transformed a number of industries, such as home automation and healthcare [1]. Infant care is one of the most important fields where technology may make a significant difference. Any instrument that helps ease this duty while guaranteeing the baby's safety and well-being is vital during the early stages of life, when a baby needs continual attention and alertness.

Babies frequently cry because they need a diaper change, are hungry, exhausted, or unwell. Since many infants who die from sudden infant death syndrome (SIDS) are discovered in cribs, the condition is sometimes known as "cot death." It affects infants younger than 12 months. Most SIDS deaths occur in babies under 6 months of age. Experts still don't know what causes SIDS, but allowing babies to sleep on a hard surface (a crib) can reduce the risk. Babies should also not sleep on pillows or other soft surfaces. Scientists don't know why sleeping in such a place increases a baby's risk of early death, but they warn that it can be dangerous. For

example, a 2003 study found that babies who slept on soft mattresses instead of hard carpets had a fivefold increased risk of SIDS. Also, avoid overheating while sleeping. Keep your baby warm while sleeping, but not too hot. The risk of SIDS increases in winter or cold weather, when parents overheat their babies by overdressing them or putting them in heavier blankets. Therefore, if the room temperature is good for adults, it will also be suitable for child.

In today's fast-paced world, working parents often struggle to balance their responsibilities with taking care of their children. After a long day at work, it can be difficult to provide the care and comfort your baby needs. Routine methods to calm the baby down, such as cuddling, may not be possible. Parents may still worry about their child's safety even after seeking help from babysitters.

This advanced technology is designed to improve the safety and health of babies while providing the necessary support to working parents. Real-time control, monitoring, as well as self-management and optimization, are the functions of the IoT. The fact that the IoT is meant to make life easier and that people can spend more money on items that do so may be one of the primary causes of its popularity.

The role of IoT in the present applications is very important as it made every application as smart. Hence IoT is incorporated into our childcare systems in this study to give parents more security and quicker reaction times. The primary microcontroller in this system is the Arduino board called Arduino Uno, which is an open platform for IoT applications. It is employed to gather sensor-read data and upload it to the server. Additionally, the GSM module receives commands from the user to carry out specific tasks. Like other development boards (such as the Raspberry Pi and Arduino), the Arduino contains a programmable circuit board. The Arduino software, which is an integrated development environment (IDE) where scripts are created and transmitted to the microcontroller, can be used to program the Arduino.

With IoT-based baby monitoring systems, devices such as smart cameras, motion sensors, temperature sensors, and wearable monitors (for tracking vital signs) are all integrated to provide real-time data on the baby's well-being [1]. These

devices, connected through IoT, offer parents and caregivers constant insights, but they also bring about several vulnerabilities:

- Vulnerabilities in encryption and access control: Baby monitoring systems that transmit sensitive data, like video feeds or health metrics, over the internet can be susceptible to hacking if encryption protocols are weak, leaving the system vulnerable to unauthorized access.
- Weak credentials and insecure applications: If the system's web or mobile application, through which parents monitor their baby, has poor password policies or lacks sufficient security, hackers can gain access, compromising both privacy and safety.
- Malwares, viruses, and worms: IoT baby monitors are potential targets for malware attacks, which could disrupt the system or give hackers control of the device, potentially leading to privacy breaches.
- Un-updated firmware and operating systems: Manufacturers of these monitoring devices often prioritize affordability and ease of use, but this can come at the expense of not regularly updating firmware or operating systems, making the system vulnerable to newly discovered threats.
- Limited computational capacity: Baby monitoring systems often use lightweight hardware (like Arduino) that is designed to be low-power and cost-effective. While this improves usability, it may also limit the system's ability to implement more advanced or resource-intensive security measures.

The scope of the baby monitoring system using IoT encompasses both technical and practical aspects. From a technical standpoint, the system involves the design, development, and implementation of a comprehensive system comprising various hardware components such as Arduino, sensors (temperature and humidity), GSM module, and a virtual terminal, along with software components for data processing, analysis, and remote monitoring. The system also includes testing and evaluation to ensure the reliability, accuracy, and effectiveness of the system in real world scenarios. Practically, the scope extends to addressing the critical need for enhanced infant safety and monitoring solutions in modern households. By providing caregivers with real-time monitoring capabilities and remote access to vital information about the baby's environment, the system aims to offer peace of mind and reassurance to parents while enabling them to respond promptly to any potential risks or emergencies.

Moreover, the scalability and customization features of the system allow for adaptation to different caregiving contexts and preferences, catering to diverse needs and lifestyles. In terms of social impact, the system addresses a fundamental concern for the well-being of infants and young children, which is of paramount importance to families and communities worldwide. By empowering parents and caregivers with advanced monitoring capabilities, the system contributes to promoting child safety and fostering a supportive caregiving environment. Additionally, the system may have economic implications by potentially reducing

healthcare costs associated with preventable infant health issues through early detection and intervention. From an environmental perspective, while the direct impact of the system may be minimal, its indirect benefits lie in promoting sustainable caregiving practices.

The advent of modern parenting has brought with it many conveniences, but also many concerns, especially about the safety and health of babies. Traditional childcare systems, while somewhat effective, do not offer solutions that meet the changing needs of today's caregivers.

## II. RELATED WORKS

One of the main problems is the limited monitoring capacity of traditional systems. While basic audio and video monitoring is useful, it often fails to capture environmental conditions such as temperature and humidity, which are important for ensuring the safety and comfort of babies. The deficiency leaves caregivers in a constant state of confusion, unable to correctly identify and respond to risks or emergencies. This image shows a caregiver who is stressed, anxious, and struggling to balance work and baby care due to lack of appropriate care.

In addition, the lack of remote access poses a serious problem for caregivers, especially those who travel and work in many ways, in today's fast-paced lifestyle. It's everywhere. Traditional childcare is often attached to or located near the child's room, limiting the caregiver's ability to provide care when the child is away from home or engaged in other activities. This restriction not only provides peace of mind for the caregiver, but also raises concerns about the child's safety in their absence. Therefore, there is an urgent need to evaluate solutions that provide access to conflict zones and allow caregivers to stay in touch with their child's well-being, no matter where they are. In addition, current monitoring systems often lack an alarm system, making it easy for supervisors to miss an alarm or delay an alert. Rapid detection of abnormalities, such as sudden changes in temperature or humidity, is important to ensure timely intervention and to reduce risks to the child's health and safety.

However, traditional systems cannot provide timely warnings or customize the warnings according to the observer's interests. This deficiency both deteriorates the quality of care and increases the caregivers' trust in the body's reliability, causing them to be anxious and uncertain about the baby's health. These are the problems faced in existing traditional baby monitoring system; therefore, new analyses are urgently needed to solve gap of existing system. And also give solutions to the problems faced by the caretaker. The complete details of referred papers and comparison of our proposed system is shown in the table1 at the end.

The following are the few research gaps identified from the existing system:

- Manual monitoring capabilities should replace with the new technologies like IoT, AI for parameters like temperature, Humidity, heartrate etc.

- Many systems depend on the sensors and algorithms, which should be properly check for reliability and accuracy.
- Wi-Fi or GSM module should be used to send alerts and notifications for caretaker or to the parents.
- Privacy and security issues should be taken care through the smart system.

### III . SMART CRADLE SYSTEM

The Child Care Center using IoT is a system designed to meet the changing needs of today's caregivers by providing effective solutions for safe care and baby health. The system combines technologies such as Arduino, GSM modules, temperature and humidity sensors, virtual terminals and LCD displays to create a monitoring system with better functionality.

Main features:

1. Environmental Monitoring: The system includes sensors to monitor important environmental factors such as temperature and humidity in the child's room. This ensures that the child is in a good and safe environment and allows caregivers to be immediately alerted to any deviations in quality.
2. Remote access: Thanks to the integrated GSM module, caregivers can access information in real time and receive notifications via their mobile devices. This gives caregivers peace of mind by allowing them to stay connected to their child's health even when they are away.
3. Customizable alarms: The system offers customizable alarms, allowing caregivers to set individual temperature and humidity levels. This ensures that supervisors receive timely alerts when conditions deviate from expectations, enabling timely intervention and risk reduction.
4. User-friendly interface: The system uses a virtual terminal and LCD screen for user interaction and provides a user-friendly interface for monitoring and control. Caregivers can easily access and interpret data in real time to inform decisions and quality care.

Figure 1 shows the initial monitoring of the baby's temperature, pulse rate and humidity. The results are displayed on the LCD screen and the system checks if it exceeds the threshold. If any value is abnormal, a warning message is sent to the administrator; otherwise, the process will be terminated.

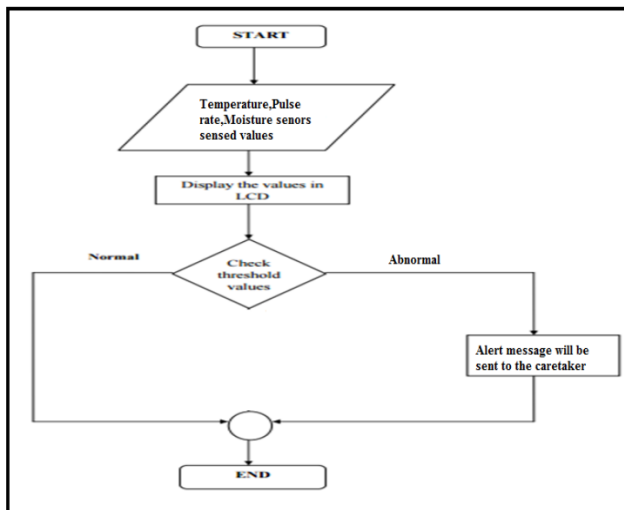


Figure 1: Block Diagram of Smart Cradle Monitoring System

### A. METHODOLOGY

The process for a baby care system involves several key stages that will ensure the development and success of the system. The Smart Cradle system architecture is shown in the figure 2.

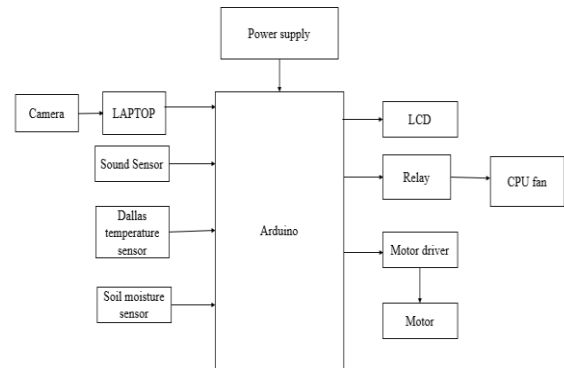
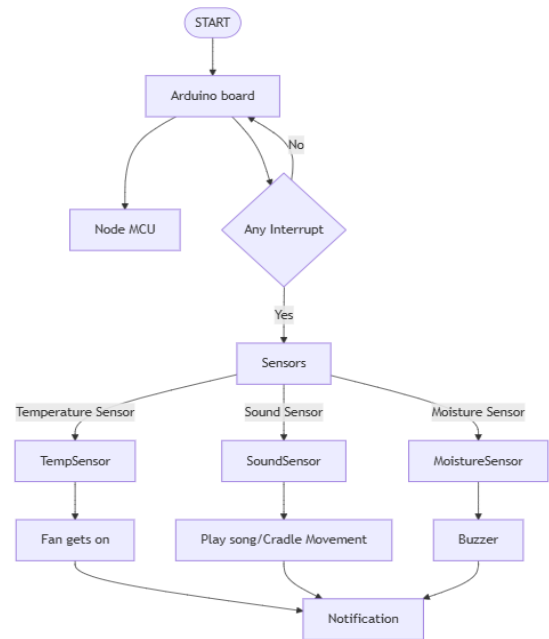


Figure 2: Architecture of Smart Cradle System

The detailed processing and monitoring is shown in figure 3, the methodology of smart cradle system as several steps, which are discussed below:

Figure 3: Process of Smart Cradle System



#### • Component Selection

In developing a reliable infant monitoring system, the components used are sensors, controllers, and communication modules. A Temperature Sensor (LM35) is used to measure ambient temperature, making sure that the surroundings stay at ease for the baby. A Humidity Sensor (normally DHT11 or DHT22) records ambient humidity levels. An Arduino Uno Board serves as the gadget's microcontroller, processing sensor data and dealing based on predefined thresholds. A Relay Module is protected to govern the Fan, which activates when the temperature exceeds a threshold level, assisting to adjust room temperature. The GSM Module (SIM900D) permits far off tracking by sending SMS notifications to caregivers in case of any atypical conditions. Moreover,

an LCD display show actual-time remarks on the systems to monitor temperature and humidity at once. Every issue is carefully selected based on capability, reliability, scalability, and simplicity of integration to make certain that the general device is powerful and meets the mission's requirements.

### • Smart Cradle Circuit Design

The circuit design is developed using the Proteus software, which permits for comprehensive simulation of the hardware components before physical meeting. Proteus permits the combination of embedded C programs for every hardware thing, which can be chargeable for particular functions, along with reading sensor information and activating the relay. The simulation allows in figuring out capability issues with the circuit format, wiring, and element interaction, ensuring that every part of the device works as anticipated. The proteus software is used to simulated the complete smart cradle system, which includes sending signals through the GSM module and updating the LCD display. In addition, blink app or Arduino IDE can be used to monitor and take actions for assistance of smart cradle system.

### • Testing and Debugging

This segment includes both unit trying out of individual additives, including the sensors, relay, and GSM module, and system integration testing to verify that all components together. The smart cradle system is thoroughly tested and gave better results.

### • Deployment

After checking out, the finalized version of the baby monitoring device is deployed in real-global settings, consisting of homes or childcare centers. Although deployment of the device's overall performance is monitored, and records is collected for the reliability and impact on caregiver behavior.

## B. IMPLEMENTATION

The smart cradle system is used to monitor the environmental conditions of the baby care and take the necessary actions to make the system is more reliable, secure, and efficient monitoring. In the existing systems where only monitoring is done, but the proposed will take the actions on the monitored data. The proposed algorithm of the baby care system is given below and the Key code snippets necessary for the system's operation are emphasized and explained. These parts are vital for ensuring the functionality and reliability of the monitoring and control system.

### ALGORITHM

#### START

Step 1: Initialize the temperature, humidity and motion sensor.

Step 2: Initialize the GSM module for sending alerts to the particular phone number.

Step 3: Connect the fan and other display devices like LCD.

Step 4: Establish the network connectivity between the IoT components for data sharing.

Step 5: Read the sensor Values

Step 6: if temperature > Threshold level  
Display and turn on FAN  
Continue

Step 7: if humidity > Threshold level  
Display and take action  
Continue

Step 8: if the sensor readings are invalid  
Display and take action  
Continue

Step9: Connect the system through wifi or GSM communicate and send alerts.

Step 10: Disconnect sensors

**STOP**

### Temperature and Humidity Reading:

This code segment is essential for obtaining data from the LM35 and DHT11 sensors, which play a central role in the system's functionality shown in figure4.

```
#include <DHT.h>
#define DHTPIN 2 // Digital pin connected to the DHT sensor
#define DHTTYPE DHT11 // DHT 11
DHT dht(DHTPIN, DHTTYPE);
int lm35Pin = A0; // Analog pin connected to LM35
void setup() {
  Serial.begin(9600);
  dht.begin();
}
void loop() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature(); // Celsius
  // Read temperature from LM35
  int lm35Reading = analogRead(lm35Pin);
  float lm35Temperature = (lm35Reading * 5.0 * 100.0) / 1024.0;
  // Print readings to Serial Monitor for debugging
  Serial.print("Humidity: ");
  Serial.print(humidity);
  Serial.print("\n Temperature (DHT11): ");
  Serial.print(temperature);
  Serial.print("\nC Temperature (LM35): ");
  Serial.print(lm35Temperature);
  Serial.println("\n");
  delay(2000); // Wait for 2 seconds before next reading
}
```

Figure 4: Code Snippet for Temperature and Humidity Reading

### Fan Control Algorithm:

This code segment manages the operation of the fan based on a specified temperature threshold. It is vital for ensuring that the fan turns on when the temperature surpasses a predetermined limit shown in figure5.

```
#define FAN_PIN 8
#define TEMP_THRESHOLD 30.0 // Temperature threshold in Celsius
void setup() {
  pinMode(FAN_PIN, OUTPUT);
  digitalWrite(FAN_PIN, LOW); // Ensure fan is off initially
}
void loop() {
  // Assume temperature variable is updated from sensor reading
  float temperature = dht.readTemperature(); // Using DHT11 sensor
  if (temperature > TEMP_THRESHOLD) {
    digitalWrite(FAN_PIN, HIGH); // Turn on the fan
    sendSMS("Fan is turned on because of high temperature: " + String(temperature) + "C");
  } else {
    digitalWrite(FAN_PIN, LOW); // Turn off the fan
    delay(2000); // Wait for 2 seconds before next check
  }
  void sendSMS(String message) {
    Serial.print("AT+CMGF=1\r"); // Set SMS mode to text
    delay(1000);
    Serial.print("AT+CMGS=\"" + 916363450449 + "\r"); // Replace with actual phone number
    delay(1000);
    Serial.print(message); // The SMS message content
    delay(1000);
    Serial.write(26); // ASCII code for CTRL+Z
    delay(1000);
  }
}
```

Figure 5: Code Snippet for Fan Control

### GSM Module Communication:

This code segment manages the interaction with the GSM module for sending SMS notifications. This functionality is essential for notifying users about changes in the system's status shown in figure6.

```
#include <SoftwareSerial.h>

SoftwareSerial gsmSerial(10, 11); // RX, TX pins for GSM module

void setup() {
  gsmSerial.begin(9600); // Start software serial for GSM module
  Serial.begin(9600); // Start hardware serial for debugging
}

void sendSMS(String message) {
  gsmSerial.print("AT+CMGF=1"); // Set SMS mode to text
  delay(1000);
  gsmSerial.print("AT+CMGS="+919763651971); // Replace with actual phone
  number
  delay(1000);
  gsmSerial.print(message); // The SMS message content
  delay(1000);
  gsmSerial.write(26); // ASCII code for CTRL-Z
  delay(1000);
}
```

Figure 6: Code Snippet for SMS notification

### LCD Display Update:

This code segment refreshes the LCD display with real-time information on temperature, humidity, and system status. This is crucial for offering users immediate feedback shown in figure7.

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(7, 8, 9, 10, 11, 12); // RS, EN, D4, D5, D6, D7

void setup() {
  lcd.begin(16, 2); // Initialize the LCD with 16 columns and 2 rows
  lcd.print("System Init...");
}

void loop() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature(); // Celsius
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(temperature);
  lcd.print("C");
  lcd.setCursor(0, 1);
  lcd.print("Hum: ");
  lcd.print(humidity);
  lcd.print("%");
  if (temperature > TEMP_THRESHOLD) {
    lcd.setCursor(10, 1);
    lcd.print("Fan ON");
  } else {
    lcd.setCursor(10, 1);
    lcd.print("Fan OFF");
  }
}
```

Figure 7: Code Snippet for display

## IV. RESULTS AND DISCUSSION

Figure 8 shows the simulation of the baby monitoring system carried out using Proteus software.

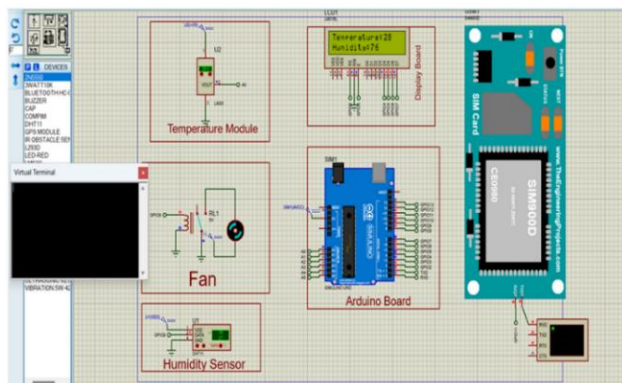


Figure 8: Simulation of Baby Monitoring System using Arduino when the temperature is around the room temperature

The simulation includes various components such as an Arduino microcontroller, GSM module, LCD display, DC motor with a fan, temperature sensor, humidity sensor, and a virtual terminal. This setup allows for comprehensive monitoring and control to ensure the baby's comfort and safety. In this scenario, the temperature sensor effectively measures the room temperature, reporting a value of up to 30 degrees Celsius. Meanwhile, the humidity sensor detects the ambient humidity levels. These readings are quickly displayed on the LCD screen, offering real-time feedback to caregivers or users about the environmental conditions in the baby's room.

The Figure 9 shows the system stimulation is observed when the temperature exceeds the predefined room temperature threshold of 30 degrees Celsius. Upon detecting a temperature surpassing this limit, the temperature sensor promptly relays this information to the Arduino microcontroller. Subsequently, the Arduino processes this data and instructs the GSM module to display a message indicating that the fan has been activated due to the high temperature. Additionally, the Arduino triggers the activation of the DC motor, which powers the fan to mitigate the elevated temperature. Concurrently, the GSM module sends a notification using the virtual terminal and displays the corresponding message on the LCD screen.

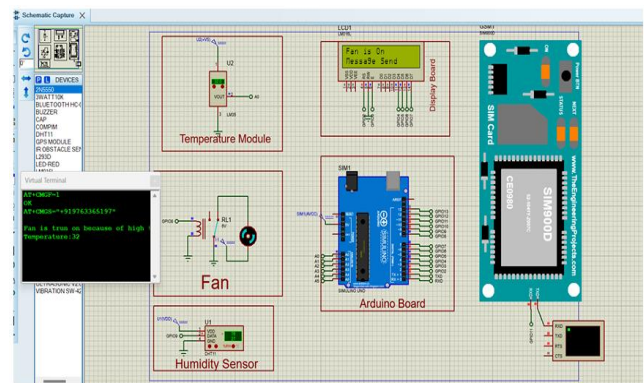


Figure 9: Simulation of Baby Monitoring System using Arduino when the temperature exceeds the room temperature

This integrated response mechanism ensures swift action is taken to regulate the room temperature and maintain a comfortable environment conducive to the baby's well-being. Through this simulation, users can witness the seamless coordination between the temperature sensor, Arduino, GSM module, DC motor, virtual terminal, and LCD display. This holistic approach demonstrates the system's capability to effectively respond to environmental fluctuations, safeguarding the baby's comfort and safety.

To confirm that the system activates the fan correctly when the temperature exceeds the specified threshold and sends an SMS notification via the virtual terminal as shown in the figure10.

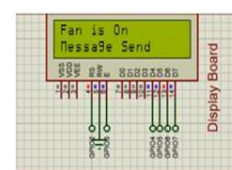


Figure 10: LCD display



To confirm that the GSM module effectively sends SMS notifications as shown in the figure 11.

1. Trigger a condition that necessitates an SMS notification (e.g., when the temperature exceeds the threshold).
2. Verify SMS reception on the designated phone number.

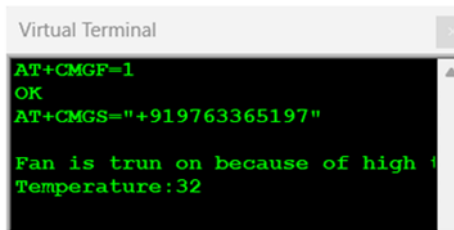


Figure11: virtual terminal

The above results are discussed for smart cradle system. Hence the comparison of Traditional Baby Monitoring System and IoT based Baby Monitoring System is given in the table 2 at the end.

## Graphical Representation of Results

### 1. Temperature vs. Time graph

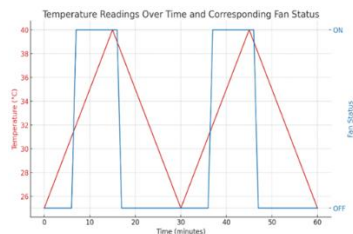


Figure 12: Temperature vs Time graph

Red Line (Temperature in °C): This line indicates the temperature readings taken over a period of time (in minutes). As the temperature increases, reaches a point where the fan needs to be activated to cool down the environment.

Blue Line (Fan Status): This line indicates the status of the fan. The fan status is represented on a secondary y-axis with "0" indicating "OFF" and "1" indicating "ON".

From the Figure 12, we can observe the following:

- The temperature starts to increase from an initial value, reaching a peak before decreasing again.
- The fan is turned ON when the temperature exceeds the threshold (around 30°C).
- The fan remains ON while the temperature stays above the threshold and turns OFF once the temperature drops below the threshold.

### 2. Humidity vs. Time

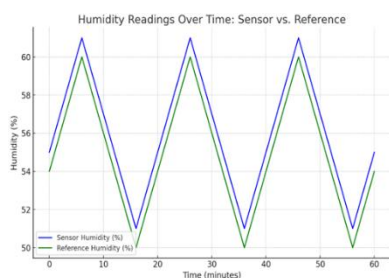


Figure 13: Humidity vs Time graph

Blue Line (Sensor Humidity in %): This line represents the humidity readings taken from the DHT11 sensor over time.

Green Line (Reference Humidity in %): This line represents the humidity readings from a reference humidity meter over the same period.

From the Figure 13, we can observe the following:

- The humidity readings from the sensor closely follow the pattern of the reference meter, demonstrating good accuracy.

The sensor readings occasionally deviate slightly but stay within a close range of the reference readings, indicating reliable performance.

### 3. SMS Notification Log:

Table 3 listing the times and contents of SMS notifications sent by the system.

Table 3: SMS notifications

| Time     | SMS Content                                        |
|----------|----------------------------------------------------|
| 10:05 AM | Fan is turned on because of high temperature: 32°C |
| 10:10 AM | Temperature is back to normal: 29°C                |

The results discussed is intended, with precise monitoring of temperature and humidity, dependable fan control, and efficient communication through SMS and the virtual terminal. The graphical representations offer a clear visualization of the system's performance over time, showcasing its responsiveness and accuracy under real-world conditions. Overall, the system evaluation demonstrates that it meets the design objectives and effectively monitors and controls environmental conditions.

## CONCLUSION

The IoT-based baby monitoring system presented in this paper offers an innovative solution to the complex challenges caregivers face in ensuring infant safety. In a time when parents often balance work with childcare, this technology effectively leverages advanced tools to enhance the caregiving environment. By incorporating components like Arduino microcontrollers, temperature and humidity sensors, and GSM modules, the system provides a comprehensive framework for continuous monitoring of critical environmental factors. This capability enables caregivers to receive real-time alerts on their mobile devices, allowing for prompt action in response to fluctuations in temperature or humidity that could affect the baby's comfort and safety. The system is designed to be both robust and user-friendly. Customizable alerts further enhance the experience, ensuring caregivers receive notifications tailored to their specific needs. This level of connectivity ensures that caregivers are not only informed but also able to respond swiftly to changes in their child's environment, ultimately promoting a safer living space for infants.

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**Table 1: The unique aspects of our paper compared to related work**

| Features/Aspect                  | Our system | [2] Patel et al. (2018) | [3] Wang et al. (2017) | [4] Joshi & Mehetre (2017) | [5] Garcia et al. (2022) | [6] Smith et al. (2023) | [7] Johnson et al. (2022) |
|----------------------------------|------------|-------------------------|------------------------|----------------------------|--------------------------|-------------------------|---------------------------|
| Multi-Sensor Integration         | Yes        | Basic sensors           | Sound detection        | Basic sensors              | Wearable sensors         | Limited to few sensors  | Advanced multi-sensors    |
| Real-Time Data Analytics         | Yes        | No                      | No                     | No                         | No                       | Basic analytics         | Yes                       |
| Customized User Profiles         | Yes        | No                      | No                     | No                         | No                       | No                      | No                        |
| Augmented Reality Integration    | Yes        | No                      | No                     | No                         | No                       | No                      | No                        |
| Voice Activation                 | Yes        | No                      | No                     | No                         | No                       | No                      | No                        |
| Telehealth Integration           | Yes        | No                      | No                     | No                         | No                       | No                      | No                        |
| Developmental Milestone Tracking | Yes        | No                      | No                     | No                         | No                       | No                      | No                        |
| Advanced Security Protocols      | Yes        | Basic security          | Basic security         | Basic security             | Limited security         | Basic security          | Basic security            |
| Energy Efficiency Features       | Yes        | No                      | No                     | No                         | No                       | No                      | No                        |
| Community Support Features       | Yes        | No                      | No                     | No                         | No                       | No                      | No                        |
| Modular Design                   | Yes        | No                      | No                     | No                         | No                       | No                      | No                        |
| Integration with Smart Devices   | Yes        | No                      | No                     | No                         | No                       | Limited integration     | No                        |

**Table 2: comparing the use of IoT in a baby monitoring system versus traditional system without IoT.**

| Features                       | IoT-Based Baby Monitoring System                                        | Traditional Baby Monitoring System                             |
|--------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------|
| Real-Time Monitoring           | Yes, continuous real-time data collection and alerts                    | Limited to manual checks, often with delayed responses         |
| Remote Access                  | Accessible via smartphones or tablets from anywhere                     | Typically, local access; requires physical presence            |
| Environmental Sensors          | Monitors temperature, humidity, and other environmental factors         | Basic audio/video monitoring only                              |
| Alert System                   | Customizable alerts for various conditions (e.g., temp changes, crying) | Basic alerts, often less customizable                          |
| Data Logging                   | Logs historical data for analysis and trends                            | No data logging capabilities                                   |
| User Interaction               | User-friendly interface with mobile notifications                       | Limited interface, often requiring physical inspection         |
| Integration with Other Devices | Seamless integration with smart home devices                            | Standalone system, limited or no integration capabilities      |
| Emergency Response             | Immediate alerts to caregivers on mobile devices                        | Delayed response time; relies on caregivers being present      |
| Scalability                    | Easily expandable to add more sensors/devices                           | Difficult to scale; often requires complete system replacement |
| Security Features              | Enhanced security with encryption and secure access                     | Basic security measures, often more vulnerable to breaches     |